

UTKARSH SINGH

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EDUCATION

SRM University Amaravati — B.Tech in Computer Science

Aug 2024 – May 2028

CGPA: **8.78/10.00** | *Coursework:* Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, System Design, Distributed Systems, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, C++, SQL, C

CS Fundamentals: DSA, System Design, OOPs, Concurrency, Testing

AI/ML & Analytics: Classical ML, LangGraph, LangChain, RAG, SLMs, LLMs (GPT-4, Llama-3)

EXPERIENCE

Co-Founder & Software Engineer — ShulinTech

Apr 2026 – Present

Udyam-Registered MSME, Govt. of India | Software Development & AI Consultancy

- Engage directly with SME clients to gather requirements, translate business needs into technical solutions, and deliver production-grade software; participate in peer code reviews and manage Git-based pull request workflows within a 3-person team.
- Shipped **Sanchay** (May 2026), a mobile inventory-management platform for real estate businesses currently serving **15+ active users**, featuring a voice-input RAG pipeline for natural-language queries — reducing manual inventory tracking time by ~60%.

PROJECTS

MerchantMind — Autonomous AI Shopping & Growth Agent (Live | Demo Video | GitHub)

Python, FastAPI, Next.js, Groq,

Razorpay, PostgreSQL, Redis

- Built a multi-agent conversational-commerce platform for natural-language and voice-based shopping, enabling discovery across **200+ Bangalore stores** and **5,000+ products** with sub-**650ms** cached catalog retrieval.
- Engineered an atomic 3-phase checkout saga using PostgreSQL row-level locks, Razorpay Orders and Payment Links, idempotency keys, and HMAC-SHA256 webhook verification to prevent overselling and duplicate payments.
- Implemented deterministic budget enforcement, contextual occasion-based upselling, prompt-injection sanitization, Redis rate limiting, and live delivery tracking; validated reliability through a **151-test** security and integration suite.

Trinetra — Cross Compliance Guardian (Patent Filed) GitHub)

Python, LangGraph, FastAPI, XGBoost, PostgreSQL

- Built a production financial-document intelligence pipeline ingesting ITR filings, GST records, annual reports, and bank statements to autonomously generate a 12-page Credit Assessment Memorandum in **under 90 seconds** — replacing 2–3 weeks of manual analyst work at NBFCs.
- Designed RESTful APIs coordinating 13 concurrent agents with sub-second task scheduling for parallel multi-format financial data processing; instrumented with health-check endpoints and structured logging for observability.
- Integrated XGBoost with SHAP/LIME explainability for transparent, auditable credit risk scoring aligned with regulatory compliance requirements — enabling client-facing audit trails critical for consulting deliverables.

Aegis Forge — Real-Time Voice AI Interview Platform (Demo Video | GitHub)

Python, FastAPI, JavaScript, LiveKit, WebRTC

- Built a real-time voice AI platform achieving **850ms end-to-end latency** across speech-to-text, LLM, and text-to-speech pipelines over LiveKit WebRTC with FastAPI backend orchestration.
- Architected a 6-agent orchestration system with event-driven inter-agent communication and shared state via a singleton knowledge engine; wrote integration tests covering session lifecycle and failover paths.
- Designed anti-cheating and integrity systems using MediaPipe behavioral biometrics for real-time gaze tracking and audit-grade candidate verification.

RudraKernel — Multi-Agent RL Environment for LLM Safety (Live | GitHub)

Python, TRL, GRPO, FastAPI

- Built an OpenEnv-compatible multi-agent reinforcement-learning environment training LLMs to detect adversarial sleeper-agent attacks and epistemic cascade failures — directly relevant to **Microsoft's Responsible AI** initiatives.
- Engineered a fine-tuning pipeline with a 9-component composable reward system, deterministic replay for auditable training, and step-gated test coverage across unit, integration, and end-to-end suites.

ACHIEVEMENTS

- Top Finalist — Meta OpenEnv × PyTorch Hackathon 2026:** Built RudraKernel with reinforcement-learning-based LLM safety training.
- Runner-Up (\$1,500 Prize) — Zenith National Hackathon:** Shipped a distributed AI system under competitive time constraints.
- Top 25 Finalist — Logithon 25 @ IIT Bombay and HackFor Green Bharat @ Microsoft Gurugram,** competing against top institutions nationally.
- Top 26 Finalist — iQOO Hackathon, Chennai**
- Competitive Programming:** 150+ DSA problems on LeetCode spanning graph algorithms, dynamic programming, and system-level optimization.